

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Senior QA Automation Engineer / SDET with over **6+ years** of experience owning quality across enterprise SaaS, financial, healthcare, API-driven, and multi-module business applications from requirements analysis through production release.
- Expertise in **Playwright, TypeScript, JavaScript, C#, REST APIs, SQL, and scalable automation architecture**, developing reliable end-to-end, regression, integration, and backend validation suites for complex enterprise workflows.
- Strong manual and exploratory testing background with experience reviewing business requirements, identifying missing acceptance criteria, uncovering edge cases, assessing risk, and building comprehensive test strategies before development is complete.
- Hands-on experience integrating automated tests with **GitLab, GitLab CI/CD, Git, Docker, and continuous delivery pipelines**, improving test reliability, execution feedback, coverage, and release confidence across Agile teams.
- Experienced testing **financial and accounting-oriented workflows, regulated enterprise applications, audit-sensitive data, and multi-module SaaS platforms**, with strong attention to data accuracy, traceability, and business impact.
- Proficient in **AI-assisted testing** for generating and refining test cases, discovering edge conditions, accelerating Playwright automation, analyzing logs and defects, assessing regression impact, and validating AI-generated outputs before framework adoption.
- Quality-focused technical leader experienced performing production RCA, writing high-quality defect reports, mentoring junior QA engineers, improving reusable test architecture, and collaborating with Product Managers, Developers, Business Analysts, and QA teams.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Automation:** TypeScript, JavaScript, C#, Python, Java, SQL, Playwright, Selenium WebDriver, Cypress, NUnit, XUnit, JUnit, TestNG, Cucumber BDD.
- **Testing & QA Strategy:** Manual Testing, Exploratory Testing, Regression Testing, End-to-End Testing, Functional Testing, Integration Testing, API Testing, Risk-Based Testing, Test Planning, Test Estimation, Release Validation.
- **API & Backend Validation:** REST APIs, Postman, Swagger/OpenAPI, RESTAssured, JSON, SQL Server, PostgreSQL, MySQL, Backend Integration Testing, Data Validation, Error Handling, Contract Validation.
- **CI/CD & Engineering Tools:** GitLab, GitLab CI/CD, Git, GitHub, Jenkins, Azure DevOps, TeamCity, Docker, CI/CD Pipelines, Automated Quality Gates, Parallel Test Execution, Continuous Delivery.
- **AI-Assisted Quality Engineering:** AI Test Generation, Edge-Case Discovery, AI-Assisted Playwright Development, Requirement Analysis, Log/Defect Analysis, Regression Impact Analysis, AI Output Validation, Test Productivity Optimization.
- **Enterprise & Financial Systems:** Multi-Tenant SaaS, Enterprise Business Applications, Financial/Accounting Workflows, Audit-Driven Systems, Regulated Environments, Data Integrity, Traceability, Business-Critical Workflows.
- **Engineering Practices:** Agile/Scrum, Clean Test Architecture, Reusable Framework Design, Root Cause Analysis, Debugging, Code Reviews, Quality Standards, Mentoring, Technical Documentation, Continuous Improvement.

Test Automation Engineer

PCMI LLC (Full Time)

Oct 2024 – Present

Durham, NC, USA

- Own testing activities for new features from requirements through release, analyzing business rules, acceptance criteria, integration dependencies, and risk to define comprehensive **manual, exploratory, regression, API, and end-to-end test coverage**.
- Develop and maintain scalable automation using **Playwright, TypeScript/C#, Selenium WebDriver, REST APIs, and SQL**, building reusable test components and reliable regression suites across complex enterprise and financial workflows.
- Perform REST API and backend integration testing using Postman, JSON payloads, SQL Server, and reconciliation queries to validate service behavior, business calculations, downstream processing, and application-to-database consistency.
- Integrate automated suites with **GitLab-style CI/CD, Azure DevOps, TeamCity, Git, and Docker-based environments**, supporting continuous execution, release validation, automated quality gates, and rapid feedback to development teams.
- Apply AI-assisted testing to generate and refine test scenarios, identify missing edge cases, accelerate automation development, review requirements for gaps, and analyze logs, defects, and regression impact while validating AI-generated results before use.
- Perform root cause analysis for production and regression failures using logs, SQL, API responses, screenshots, and debugging tools while creating actionable bug reports with clear reproduction steps, severity, technical evidence, and business impact.
- Mentor QA team members through automation reviews, reusable framework patterns, test-design guidance, and process improvements while partnering with developers, Product Owners, and Business Analysts to raise overall Quality Engineering standards.

Environments: Playwright, TypeScript, C#, Selenium WebDriver, REST APIs, Postman, SQL Server, GitLab CI/CD, Azure DevOps, TeamCity, Git, Docker, AI-Assisted Testing, Financial Workflows, Exploratory Testing, RCA, Regression Testing, Agile/Scrum.

SDET / QA Automation Engineer

Client: CVS

Dec 2023 – Sept 2024

Durham, NC, USA

- Owned quality for complex healthcare and financial transaction workflows by translating user stories and business requirements into detailed manual, exploratory, integration, regression, and automated end-to-end scenarios.
- Built and maintained automation using **TypeScript, JavaScript, Playwright, Selenium WebDriver, RESTAssured, and SQL**, increasing coverage across web, API, backend, and multi-system enterprise application workflows.
- Validated REST APIs and backend integrations using **Postman, Swagger/OpenAPI, JSON, SQL, and data reconciliation**, ensuring accurate processing, error handling, data integrity, and consistent behavior across integrated services.
- Integrated automated tests with **Jenkins, GitLab/GitHub, Git, and CI/CD pipelines**, supporting continuous regression, build validation, release readiness, and timely quality feedback during two-week Agile delivery cycles.
- Used AI-assisted techniques to accelerate test design, identify edge cases, analyze failed executions, improve automated scripts, and validate generated outputs while maintaining engineering review before production adoption.
- Investigated defects and production issues using Splunk, API traces, SQL, application logs, and test evidence, performing RCA and communicating clear business impact, reproduction steps, and recommended regression coverage.
- Collaborated with Product Managers, developers, Business Analysts, and QA peers to clarify requirements, identify missing acceptance criteria, improve testability, mentor team members, and strengthen repeatable Quality Engineering practices.

Environments: TypeScript, JavaScript, Playwright, Selenium WebDriver, RESTAssured, Postman, Swagger/OpenAPI, SQL, Jenkins, GitLab/GitHub, Git, CI/CD, Splunk, AI-Assisted Testing, Enterprise SaaS, Exploratory Testing, RCA, Agile/Scrum.

Software Engineer

Infor

Sept 2021 – Dec 2022

Hyderabad, India

- Designed and executed manual, exploratory, functional, regression, integration, and end-to-end testing across enterprise business applications, identifying requirement gaps, edge cases, and release risks before feature completion.
- Developed maintainable automation using **Java, JavaScript, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL**, creating reusable framework components and scalable regression coverage across multiple application modules.
- Performed backend and API validation using Postman, SoapUI, JSON, XML, REST/SOAP services, and SQL to verify integration logic, data transformations, financial-style calculations, and downstream system consistency.
- Integrated automated suites with **Jenkins, Git, Maven, and CI/CD pipelines**, enabling continuous regression, build verification, automated execution, and dependable release validation throughout iterative development cycles.
- Executed root cause analysis for automation and application failures using logs, SQL results, API responses, and debugging techniques while documenting high-quality defects and expanding regression protection around identified issues.
- Improved framework reliability by refactoring reusable components, stabilizing brittle tests, enhancing test data handling, and increasing maintainability across growing regression suites and interconnected enterprise workflows.
- Collaborated with developers, product teams, analysts, and QA engineers to review requirements, improve acceptance criteria, share automation standards, mentor peers, and continuously evolve Quality Engineering processes.

Environments: Java, JavaScript, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, Postman, SoapUI, REST APIs, SQL, Jenkins, Git, Maven, CI/CD, Enterprise Applications, Exploratory Testing, Framework Architecture, RCA, Agile/Scrum.

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Executed comprehensive manual and automated testing across enterprise web, API, database, and integration workflows, covering functional, exploratory, regression, end-to-end, and negative test scenarios throughout Agile releases.
- Developed reusable automated test suites using **Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL**, improving repeatability and regression coverage for business-critical application functionality.
- Performed API and backend testing using **Postman, SoapUI, REST/SOAP services, JSON, XML, and SQL**, validating request-response behavior, business rules, error handling, and data integrity across connected systems.
- Reviewed requirements and acceptance criteria with developers and analysts to identify missing scenarios, boundary conditions, integration risks, and high-impact user paths before implementation reached final validation stages.
- Integrated regression suites with **Jenkins, Git, and CI/CD pipelines**, supporting repeatable automated execution, build-level feedback, release validation, and earlier discovery of defects during development.
- Investigated failed tests and application defects using logs, database queries, API responses, screenshots, and debugging evidence while documenting reproducible defects and verifying fixes through targeted regression testing.
- Contributed to reusable QA processes, test documentation, automation standards, code reviews, peer guidance, and continuous improvement initiatives that strengthened reliability and maintainability across enterprise software releases.

Environments: Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, Postman, SoapUI, REST/SOAP APIs, SQL, Jenkins, Git, CI/CD, Manual/Exploratory Testing, Regression Testing, Enterprise Applications, Defect Management, RCA, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.